

Let's Encrypt HTTPS — Status Summary

Revision: 2 **Last modified:** 2026-07-01T10:06:00Z **Status:**
Companion summary of [Status.md](#) (§11.4.56 two-audience).

Page 1 — For the operator / stakeholders (plain language)

We are adding **automatic HTTPS** to the proxy using **Let's Encrypt** — free, auto-renewing TLS certificates — with **Caddy** doing the certificate work in-process (no scripts or cron to babysit) and the **DNS-01** method so it works even without opening ports on the host.

What works today:

- The three setup decisions are locked in (Caddy, DNS-01, test-first then defer the real domain).
- A **certificate checker** is built and proven: it can tell whether a resulting certificate is still valid, how many days are left, whether it covers the right hostname, and whether it was issued by the expected authority — all offline, and it is self-tested so it cannot lie (a deliberately-broken certificate is always caught).
- A **real certificate is now issued end-to-end, fully on this machine** (no internet, no real domain): Caddy asks a local test certificate-authority for a certificate using the DNS method, gets a genuine one, and the checker confirms it is valid, covers the right name, and was really signed by that authority. This runs from scratch on demand in about 15 seconds and produces saved proof each time.

What's still pending / needs you:

- **Renewal + rotation** (proving the certificate auto-renews with no downtime) — the next automated step; the research for it is done.
- **Going live on the real domain** needs you to provide a DNS provider API token and give the go-ahead. Until then nothing touches production.

Bottom line: the foundation, the safety-checker, AND real end-to-end certificate issuance are done and proven on this machine; renewal/rotation is next and the real-domain go-live waits on your go-ahead.

Page 2 — For software engineers

- **Task:** #59 Let's Encrypt HTTPS + auto-renewal + rotation (feature workstream, §11.4.167).
- **Phase 0 (PASS):** operator decisions captured via §11.4.66 — Caddy auto-HTTPS / DNS-01 / hermetic+staging-first, defer real domain. Design LETSENCRYPT_HTTPS.md + plan_PLAN.md @ fddd25c.
- **Phase 1 cert-analyzer (PASS):** tests/letsencrypt/cert_analyzer.sh — 5 pure, now-seam-deterministic (§11.4.50 CERT_ANALYZER_NOW_EPOCH) functions; SAN match is exact-token + single-leading-wildcard (no substring, mirrors_code_in); chain check is issuance-only (openssl verify -no_check_time), orthogonal to expiry. Self-validated §11.4.107(10): cert_analyzer_selftest.sh 37/37; tests/regression/cert_analyzer_selfvalidation_test.sh GREEN+RED_MODE=1 PASS; §1.1 mutation (SAN exact→substring) → guard FAIL, restore byte-identical md5 a06e0f89ee9993fe5de368be852e4eff. Fixtures: public *.pem tracked, *.key/*.csr/*.srl gitignored (§11.4.10), regen via fixtures/gen_fixtures.sh (§11.4.77).
- **Phase 1 custom Caddy image (PASS):** deploy/letsencrypt/build.sh builds localhost/helix_proxy/caddy-challtestsrv:2.8.4 (rootless, non-root user + cap_net_bind_service) embedding the local dns.providers.challtestsrv libdns module via xcaddy; post-build anti-bluff caddy list-modules | grep dns.providers.challtestsrv PASS (§11.4.6 — build ≠ linked).
- **Phase 2 compose + CoreDNS SOA front (PASS):** compose.hermetic.yml — pebble (PEBBLE_VA_ALWAYS_VALID=0) + challtestsrv + **coredns** + caddy; secrets NAME/PATH-only, commented for the hermetic path; caddy loopback-bound (security-audit M1). CoreDNS answers hermetic.test SOA in the ANSWER section (certmagic's zone-determination needs it — challtestsrv NOTIMPs SOA) and fall-through-forwards the dynamic_acme-challenge TXT to challtestsrv.
- **Phase 3 hermetic issuance (PASS):** deploy/letsencrypt/phase3_hermetic_issue.sh — clean-slate, re-runnable (§11.4.98): boots the stack, Caddy obtains a REAL cert via DNS-01 from local Pebble (genuine validation), cert-analyzer verifies not_expired + SAN + negative-SAN-reject + chain-to-this-run's-Pebble-CA. Two clean runs, evidence in qa-results/letsencrypt/phase3_issuance/. Root-cause research for the zone-determination + a certmagic #354 panic in docs/research/letsencrypt_hermetic_20260701/ + docs/research/certmagic_chain_panic_20260701/.

- **Pending (autonomous):** Phase 5 renewal/rotation sim (research done — docs/research/letsencrypt_renewal_20260701/: renewal_window_ratio 1 via admin /load).
- **OPERATOR-BLOCKED:** Phase 4 LE-staging (real DNS-01 token §11.4.10), Phase 6 prod cutover (real-domain go-live gate, design §9).
- **Guard wiring:** cert_analyzer_selfvalidation_test.sh registered; the Phase-3 issuance guard (tests/letsencrypt/phase3_issuance_guard.sh) + LE Challenge land with this lane (§11.4.135).
- **Anti-bluff:** every non-PASS row is honestly PENDING/ OPERATOR-BLOCKED (§11.4.6) — no metadata-only PASS; every PASS cites a captured-evidence path.